UDP Log Viewer

Cross-platform UDP log viewer for ESP32 and other embedded systems, built with Python and PyQt5.

Why using this tool?

Debugging embedded devices over UDP is painful. This tool makes it simple and visual. Not only viewing data, visualizing data as well.

Documentation

Complete English documentation:

- docs/DOCUMENTATION_en.md
- docs/USER_GUIDE_en.md
- docs/RELEASE_0.16.2.md

Complete German documentation:

- docs/DOKUMENTATION_de.md

HowTo use UDP on EPS32 Demo project

please take a look here: https://github.com/mrRobot62/esp-udp-logging.git

Windows-User

please use setup.exe to install this viewer

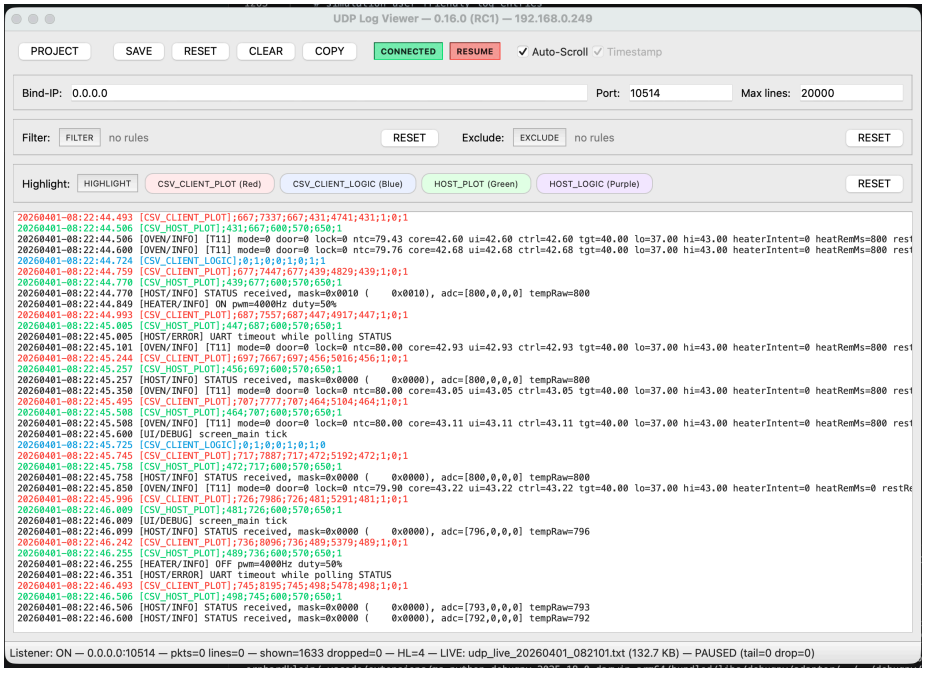
ToDo for MAC users

after installation this application is not signed and can't be started directly. You have to set this command: `xattr -dr com.apple.quarantine /Applications/UDPLogViewer.app`

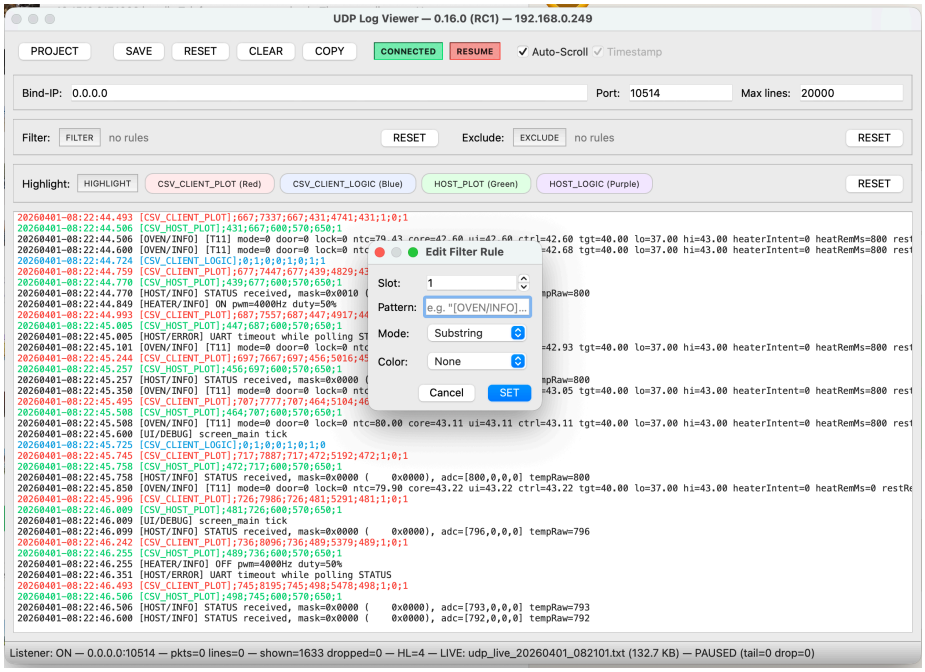
Screenshots

Screenshot assets are stored under assets/screenshots.

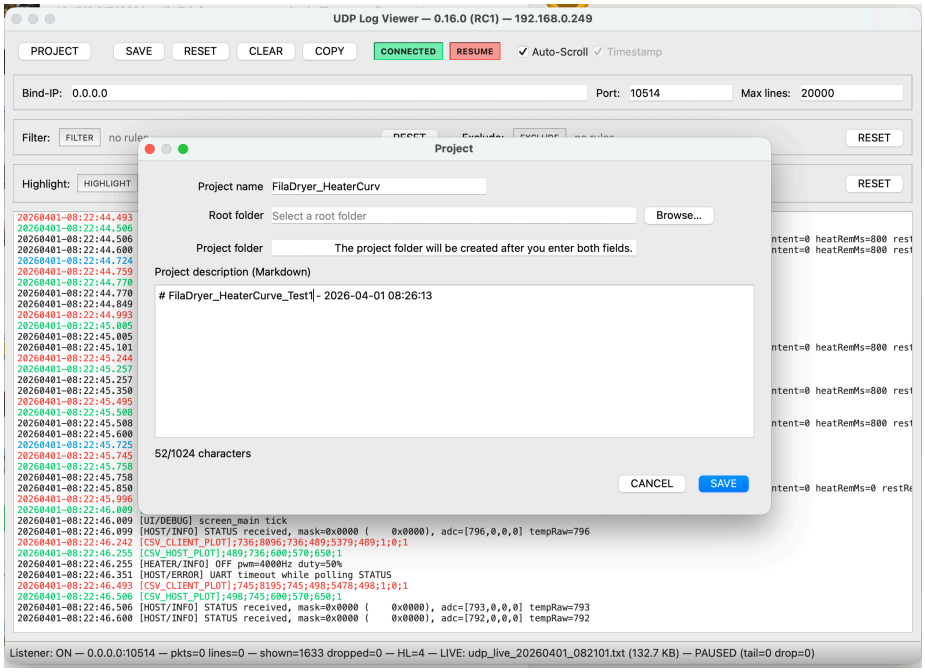
Main Screen


UDP Log Viewer — 0.16.0 (RC1) — 192.168.0.249
PROJECT
SAVE
RESET
CLEAR
COPY
CONNECTED
RESUME
Auto-Scroll
Timestamp
Bind-IP: 0.0.0.0
Port: 10514
Max lines: 20000
Filter:
FILTER
no rules
RESET
Exclude:
EXCLUDE
no rules
RESET
Highlight:
HIGHLIGHT
CSV_CLIENT_PLOT (Red)
CSV_CLIENT_LOGIC (Blue)
HOST_PLOT (Green)
HOST_LOGIC (Purple)
RESET
20260401-08:22:44.493 [CSV_CLIENT_PLOT];667;7337;667;431;4741;431;1;0;1
20260401-08:22:44.506 [CSV_HOST_PLOT];431;667;600;570;650;1
20260401-08:22:44.506 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=79.43 core=42.60 ui=42.60 ctrl=42.60 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=800 rest
20260401-08:22:44.600 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=79.76 core=42.68 ui=42.68 ctrl=42.68 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=800 rest
20260401-08:22:44.724 [CSV_CLIENT_LOGIC];0;1;0;0;1;0;1;1
20260401-08:22:44.759 [CSV_CLIENT_PLOT];677;7447;677;439;4829;439;1;0;1
20260401-08:22:44.770 [CSV_HOST_PLOT];439;677;600;570;650;1
20260401-08:22:44.770 [HOST/INFO] STATUS received, mask=0x0010 (0x0010), adc=[800,0,0,0] tempRaw=800
20260401-08:22:44.849 [HEATER/INFO] ON pwm=4000Hz duty=50%
20260401-08:22:44.993 [CSV_CLIENT_PLOT];687;7557;687;447;4917;447;1;0;1
20260401-08:22:45.005 [CSV_HOST_PLOT];447;687;600;570;650;1
20260401-08:22:45.005 [HOST/ERROR] UART timeout while polling STATUS
20260401-08:22:45.101 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=80.00 core=42.93 ui=42.93 ctrl=42.93 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=800 rest
20260401-08:22:45.244 [CSV_CLIENT_PLOT];697;7667;697;456;5016;456;1;0;1
20260401-08:22:45.257 [CSV_HOST_PLOT];456;697;600;570;650;1
20260401-08:22:45.257 [HOST/INFO] STATUS received, mask=0x0000 (0x0000), adc=[800,0,0,0] tempRaw=800
20260401-08:22:45.350 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=80.00 core=43.05 ui=43.05 ctrl=43.05 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=800 rest
20260401-08:22:45.495 [CSV_CLIENT_PLOT];707;7777;707;464;5104;464;1;0;1
20260401-08:22:45.508 [CSV_HOST_PLOT];464;707;600;570;650;1
20260401-08:22:45.508 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=80.00 core=43.11 ui=43.11 ctrl=43.11 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=800 rest
20260401-08:22:45.600 [UI/DEBUG] screen_main tick
20260401-08:22:45.725 [CSV_CLIENT_LOGIC];0;1;0;0;1;0;1;0
20260401-08:22:45.745 [CSV_CLIENT_PLOT];717;7887;717;472;5192;472;1;0;1
20260401-08:22:45.758 [CSV_HOST_PLOT];472;717;600;570;650;1
20260401-08:22:45.758 [HOST/INFO] STATUS received, mask=0x0000 (0x0000), adc=[800,0,0,0] tempRaw=800
20260401-08:22:45.850 [OVEN/INFO] [T11] mode=0 door=0 lock=0 ntc=79.90 core=43.22 ui=43.22 ctrl=43.22 tgt=40.00 lo=37.00 hi=43.00 heaterIntent=0 heatRemMs=0 restRe
20260401-08:22:45.996 [CSV_CLIENT_PLOT];726;7986;726;481;5291;481;1;0;1
20260401-08:22:46.009 [CSV_HOST_PLOT];481;726;600;570;650;1
20260401-08:22:46.009 [UI/DEBUG] screen_main tick
20260401-08:22:46.099 [HOST/INFO] STATUS received, mask=0x0000 (0x0000), adc=[796,0,0,0] tempRaw=796
20260401-08:22:46.242 [CSV_CLIENT_PLOT];736;8096;736;489;5379;489;1;0;1
20260401-08:22:46.255 [CSV_HOST_PLOT];489;736;600;570;650;1
20260401-08:22:46.255 [HEATER/INFO] OFF pwm=4000Hz duty=50%
20260401-08:22:46.351 [HOST/ERROR] UART timeout while polling STATUS
20260401-08:22:46.493 [CSV_CLIENT_PLOT];745;8195;745;498;5478;498;1;0;1
20260401-08:22:46.506 [CSV_HOST_PLOT];498;745;600;570;650;1
20260401-08:22:46.506 [HOST/INFO] STATUS received, mask=0x0000 (0x0000), adc=[793,0,0,0] tempRaw=793
20260401-08:22:46.600 [HOST/INFO] STATUS received, mask=0x0000 (0x0000), adc=[792,0,0,0] tempRaw=792
Listener: ON — 0.0.0.0:10514 — pkts=0 lines=0 — shown=1633 dropped=0 — HL=4 — LIVE: udp_live_20260401_082101.txt (132.7 KB) — PAUSED (tail=0 drop=0)

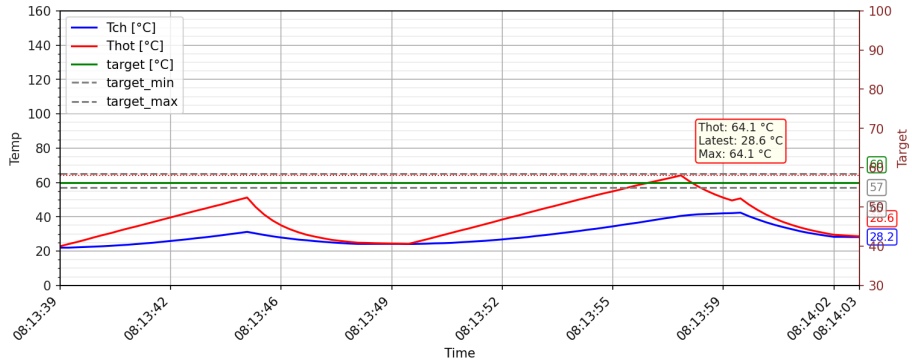
Rule Configuration



Project Dialog

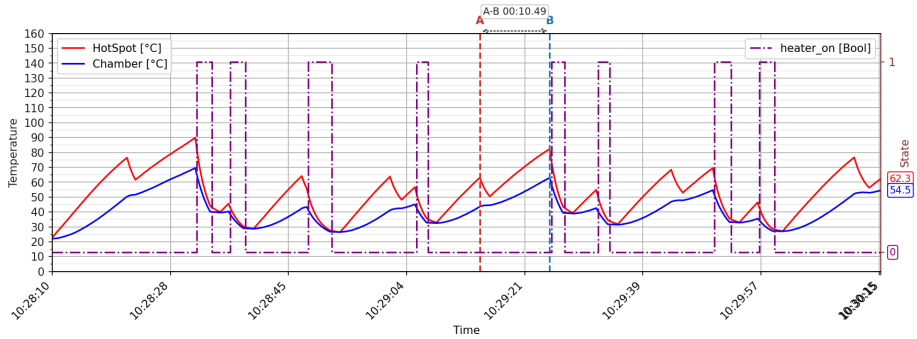


Plot Visualizer Tooltip

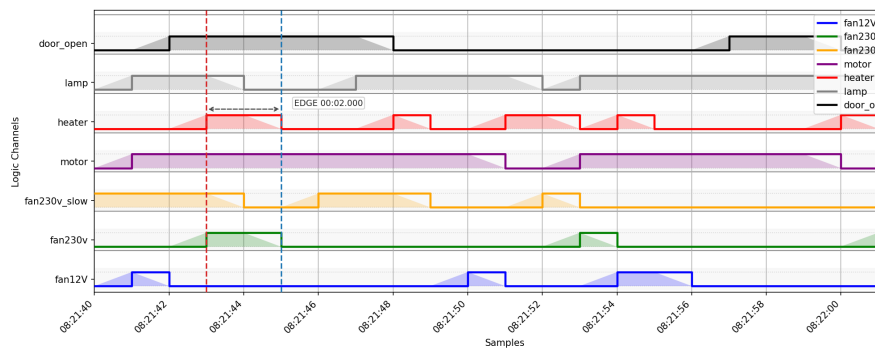


Timebased visualizing

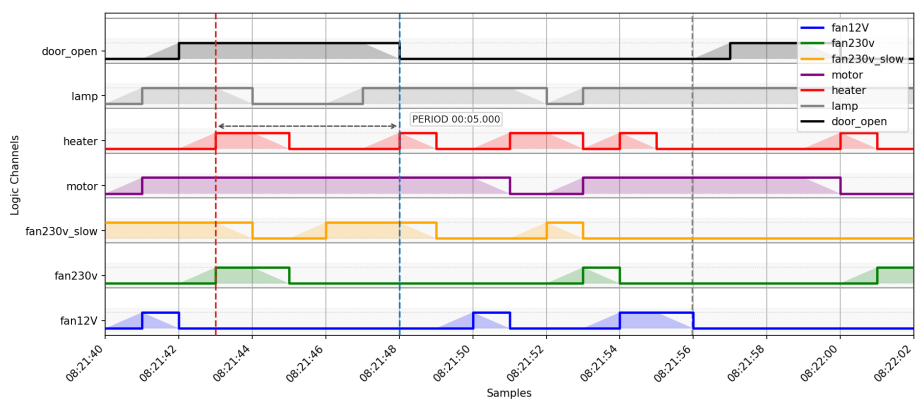
Measuring time



Logic Measurement



Measuring HIGH/LOW phase



Measuring signal period

Current Scope

The current codebase includes:

- real-time UDP log reception
- filter, exclude, and highlight rules
- live session logging

- main-window `RESET` support to start a fresh log phase inside the same app session
- runtime project contexts for grouping logs and screenshots per test session
- project descriptions stored as `README_<projectname>.md` inside the project folder
- replay of saved log files
- built-in simulation for text, temperature, and logic traffic
- CSV-based data visualization
- logic-channel visualization
- edge and period measurement directly inside the logic graph
- runtime `Legend` toggle in plot and logic graph windows
- keyboard-driven screenshot/save shortcuts and explicit `TAB` navigation in the main and graph windows
- macOS and Windows packaging scripts

Run From Source

```
python -m venv venv
source venv/bin/activate
pip install -e .[dev]
udp-log-viewer
```

Developer Bootstrap

```
./scripts/bootstrap_macos_linux.sh
```

You like it?

If you like this project, consider giving it a star!